Innovations in LMIC Neurosurgical Care: A Landscape Overview

This document provides a brief survey of existing and emerging innovations across the TBI care continuum in low-resource settings. It is intended as a starting point for research and inspiration, not a constraint on what teams can propose.

Diagnosis Without CT

The absence of CT imaging is one of the most critical barriers in LMIC TBI care. Several technologies are being developed or deployed to fill this gap:

- Near-infrared spectroscopy (NIRS): The Infrascanner 2000 is a handheld, portable device that uses near-infrared light to detect intracranial hematomas. In a study of 384 TBI patients in Uganda, it achieved 88.9% sensitivity for detecting intracranial bleeds though specificity was lower (37.7%), particularly in patients with darker skin tones. It costs a fraction of a CT scanner and requires minimal training.

- Point-of-care ultrasound (POCUS): Transcranial Doppler and B-mode ultrasound can non-invasively assess raised intracranial pressure, cerebral blood flow, and even detect intracranial hematomas at the bedside. Competence is achievable without formal radiology training.

- Smartphone-based ultrasound: In Cameroon, a 5-hour training curriculum enabled trauma providers to achieve eFAST competence using smartphone-connected ultrasound probes. 100% of trained providers met competency benchmarks.

- Portable low-field MRI: Commercially available portable MRI units (e.g., Hyperfine Swoop) have been deployed in Pakistan, Malawi, and other LMICs. They require no shielded room and can operate in standard clinical environments, providing brain imaging where CT is unavailable.

Surgical Task-Shifting

- Indonesia's national program (launched late 2025): A government-endorsed initiative trains selected general surgeons to perform emergency craniotomy for epidural and subdural hematomas under real-time neurosurgical tele-proctoring. Training uses a three-phase pathway: VR simulation, 3D-printed skull models, supervised operative experience, then independent practice with remote oversight.

- Tanzania's train-forward model: A pioneering program trained local healthcare workers to perform basic and emergency neurosurgical procedures, then taught them to train others. Over 5 years, independent case performance rose from 44% to 86%, and postoperative complications decreased significantly.

- Australia's rural craniotomy guidelines: A tertiary neurosurgical unit published formal recommendations for non-neurosurgeons performing emergency craniotomy in rural hospitals, including diagnostic pathways with and without CT access.

- A scoping review of over 2,000 emergency neurosurgical procedures performed by non-neurosurgeons across 13 countries found that emergency decompression may be lifesaving, with mortality consistently lower for epidural hematomas than subdural hematomas.

Simulation and Training Tools

- 3D-printed skull models: Low-cost 3D-printed skulls (as little as $145/participant for a full bootcamp) have been validated for practicing burr holes and craniotomies. Entry-level desktop printers can produce multi-material models that neurosurgeons rate as realistic for training.

- Virtual reality simulation: VR platforms allow trainees to practice cranial procedures in zero-risk environments. AI-powered "Virtual Operative Assistants" can classify skill level and provide automated feedback. Indonesia's task-sharing program integrates VR as a core training component.

- Low-cost exoscopes: A simplified exoscope built from an industrial microscope tube, LED ring light, and digital camera provides magnification and illumination comparable to conventional microscopes — total cost approximately $750.

Telemedicine and Remote Guidance

- Tele-neurosurgery networks have reduced time-to-specialist evaluation from ~160 minutes to ~38 minutes and prevented up to 44% of unnecessary patient transfers.

- In the Philippines, a telemedicine program for socioeconomically disadvantaged neurosurgical patients achieved a benefit-cost ratio of 3.51, with 64% of patients reporting that telemedicine prevented them from borrowing money to access care.

- Key barriers remain: connectivity, imaging interoperability, and digital inequality, particularly relevant in rural settings like the one described in this case.

Rehabilitation in Low-Resource Settings

- Mobile health (mHealth): Smartphone-based tools for memory compensation, ecological momentary assessment, and self-management have shown promise in TBI rehabilitation. mHealth is low-cost, does not require clinic access, and can overcome geographic barriers, but evidence from LMICs remains very limited.

- Community health worker models: Task-shifting rehabilitation to community-based providers is a recognized strategy, though formal TBI-specific programs in LMICs are scarce. Early neurorehabilitation guidelines for Africa have been proposed but not yet widely implemented.

- Intraoperative ultrasound for follow-up: In Tanzania, neurosurgeons have adopted low-cost intraoperative ultrasound for surgical navigation and postoperative assessment as an affordable, locally available alternative to repeat CT imaging.

Locally Adapted Surgical Tools

In rural Tanzania, neurosurgeons have adapted commonplace local items to substitute for specialized instruments, demonstrating that context-specific engineering using locally available materials can support basic neurosurgery without dependence on imported equipment. Community-specific engineering programs that produce tools locally also stimulate the local economy and ensure replacement and repair are feasible.

Systems and Infrastructure

- Electronic medical records and standardized clinical protocols have been independently associated with increased neurosurgical capacity and reduced post-TBI mortality in LMICs, suggesting that even basic systems-level interventions can improve outcomes.

- Equipment donation programs (e.g., through the World Federation of Neurosurgical Societies) have expanded surgical capacity, though high-maintenance items like high-speed drills have a 100% breakage rate, highlighting the need for durable, low-maintenance designs.